
Paper 020: From Failed Attribution to Local Mechanism Assays Residual Transport and Hash Independence on the OPHIS Plateau

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Abstract

The preceding block did not establish either of its proposed mechanisms. Its boundary-sidecar diagnostic was invalid because the scanner/scanner null diverged in model, optimizer, and loss state before treatment activation; its canonicalized n-gram keys were killed by their frozen offline mediator. We therefore restart from the unchanged current recipe and preregister at most five mechanism questions, linked by a single thesis: useful sublayer and memory signals must remain distinguishable as they are transported into the residual stream. Three independent proposer/critic passes rank gradient-preserving activation scaling (GPAS) first, followed by balanced prime moduli, a conditional constant-parameter four-hash child, conditioned attention initialization, and a conditional attention-side Peri-RMSNORM. Every rating is printed explicitly. The top choice is plausible because the NeurIPS 2025 source isolates stop-gradient as the difference between forward downscaling that fails and gradient-preserving scaling that improves perplexity; nevertheless, OPHIS is only eight layers and already contains per-head normalization, normalized MLP output, learned residual/ x_0 paths, zero output projections, and hashed memory. A baseline applicability assay must therefore show the source mechanism locally before any endpoint is allowed. Accepted-only comparator succession, same-seed concurrent controls, mechanism mediators, compiler and throughput gates, and a $1 \rightarrow 3 \rightarrow 6 \rightarrow 10$ evidence fun-

nel are frozen. Historical experiment 501 at 0.927183 remains only the chart origin. This paper reports no new SOTA and authorizes no run until its typed records, source capsules, and setup reconciliation pass.

1. Authority, Closures, and Comparator

This paper prospectively supersedes only the unexecuted experimental authority of Paper-019 Rounds 3–5 and defines a new block of at most five mechanism questions. It neither alters nor reinterprets the two terminal Paper-019 closures (GPT Research Coordinator et al., 2026). Round 1 is INVALID_DIAGNOSTIC: its exact cross-device scanner/scanner null failed before sidecar activation, so the proposed sidecar mechanism was never tested. Round 2 is KILL_BEFORE_GPU_SCORING: the conservative affected trigram share was 3.7155%, below its frozen 10% mediator. This rejects only that keys-only canonicalization retrofit, not Engram, distinct moduli, or conditional memory generally.

Neither closure supports adoption, mechanism rejection beyond its stated scope, comparator replacement, or SOTA. The cumulative comparator begins at R_0 , the current canonical recipe bound by setup reconciliation v31. V31 differs from v30 only by the default-off GPAS implementation; R_0 explicitly uses GPAS_ENABLE=0. Its 0.934930 mean and effective $\sigma = 0.001198$ are operational setup references, not substitutes for concurrent controls. The active two- σ magnitude is

$$2\sigma = 0.002396.$$

Every causal estimand uses a fresh same-seed concurrent control on the current recipe. Historical experiment 501 at 0.927183 remains the chart origin and a provenance observation. It is forbidden as a control, comparator qualification target, effect denominator, arm configuration, or SOTA claim.

The prior diagnostic’s source hashes were bound, but

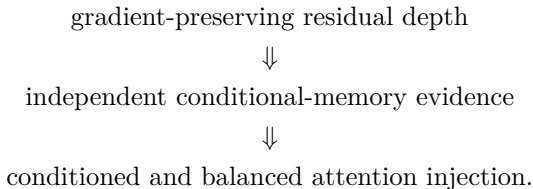
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Internal OPHIS preregistration in ICML two-column format. Not peer reviewed, submitted to, or accepted by ICML.

both RunRecords state `git_dirty=true`; we therefore do not describe it as a clean-tree execution. Its cross-device exact-state criterion is also not reused: `train.py` explicitly warns that the fixed-seed CUDA pipeline is not bitwise deterministic. Future zero-effect equivalence is performed as same-physical-device replay before an arm contrast.

2. Connected Thesis and Attribution Rule

The model already combines full in-document attention with token and hashed n-gram values. Its plateau can therefore arise even when useful features exist: a residual shortcut may dominate deeper transformations, multiple hash heads may alias the same contexts together, or an attention branch may begin in a poorly conditioned or badly scaled state. The proposed body is



This is a connected cumulative program, not permission to bundle additions. Each treatment differs from its current comparator by exactly one mechanism. A mechanism joins the comparator only after the full pre-registered acceptance funnel, not after an offline pass, a diagnostic, or one paired screen. Rejection, invalidity, or a failed mediator leaves the comparator unchanged. A conditional child cannot execute unless its named parent was accepted.

The governing principle is:

Before a system can improve itself, it must be able to observe and attribute its failures.

Consequently, an observed endpoint without its predicted mediator is evidence against the stated explanation, even when numerically favorable. It is not silently relabeled as a success.

3. Multiagent Ranking and Selection

All vectors use novelty (N), primary-source provenance (P), causal validity (V), expected local impact (I), transfer reliability (R), feasibility/cost (F), falsifiability (X), and connected-program coherence (C), each on a 1–5 scale. Institutional or conference prestige can strengthen provenance; it cannot establish local transfer. Table 1 records both proposer and independent critic scores rather than only a total.

Why GPAS is ranked first. An initial portfolio critic scored GPAS $4/5/4/3/3/5/5/5 = 34/40$. The deeper source/code transfer audit lowered reliability by one point, producing the authoritative $33/40$ shown above. GPAS makes an unusually sharp causal prediction. In a Pre-LN residual sublayer with post-sum activation z , it applies

$$z' = z - \text{SiLU}(\alpha_\ell) \text{sg}(z).$$

The forward scale is $1 - \text{SiLU}(\alpha_\ell)$, while $\partial z'/\partial z = I$. The official implementation uses one zero-initialized scalar per layer and reuses it after both the attention and FFN residual sums (Chen et al., 2025; 2026). Across 71M, 130M, 250M, 350M, and 1B Pre-LN models, the paper reports evaluation perplexity changes $33.98 \rightarrow 33.38$, $26.61 \rightarrow 26.25$, $21.54 \rightarrow 21.34$, $20.71 \rightarrow 19.77$, and $16.53 \rightarrow 16.11$. More importantly for attribution, the 350M ablation reports 20.35 with stop-gradient, 21.34 without it, and 21.35 with no GPAS, while forward variance is still reduced without stop-gradient. This isolates gradient preservation from mere downscaling.

The selection is still a transfer hypothesis, not a replication claim. The source reports no error bars or multi-seed significance, and its models are 12 layers or deeper. OPHIS has eight layers, learned residual and x_0 paths, per-head Q/K normalization and gating, normalized MLP output, zero-initialized c_{proj} weights, Muon/AdamW/RMSProp optimizer partitioning, and hashed value memory. Those differences lower the critic’s impact and reliability scores and make the baseline applicability mediator mandatory.

4. Round 1: Source-Faithful GPAS

Intervention. Add one parameter $\alpha_\ell = 0$ for each of eight layers. Reuse that scalar after both the attention and MLP residual sums, exactly as in the official implementation. The gate joins the same local AdamW class used for learned residual scalars: paper-frozen learning rate 0.005, betas (0.8, 0.95), epsilon 10^{-10} , no weight decay, and no momentum-decay flag. This value is a constant, not a derivation from current R_0 ’s `scalar_lr= 0.8` (the ratio is 0.00625). This optimizer mapping is an OPHIS-specific extrapolation; the source uses Adam at 5×10^{-4} for all parameters. This wording was corrected prospectively before any Round-1 GPU execution or result.

Zero-effect replay. Before a mechanism assay, one physical H200 runs same-device control replay and an $\alpha = 0$ treatment replay from separately constructed but source-identical states. At step 0, logits, loss, all pre-existing parameter gradients, data

Table 1. Prospective ratings in N/P/V/I/R/F/X/C order. The implementation audit found a stale width and a non-reproducible serialized definition in the first prime-modulus draft; Round 2 therefore remains blocked until a separately versioned corrected authority passes exact equality checks.

candidate	proposer	independent critic	verdict
source-faithful GPAS shared residual gate	4/5/5/4/4/5/5/5 = 37	4/5/4/3/2/5/5/5 = 33	rank 1
distinct balanced prime moduli for raw-ID	4/5/5/3/3/5/5/5 = 35	4/4/2/2/2/3/2/5 = 24	hold / blocked
n-gram heads			
source-faithful conditioned Q/K/V initialization	4/5/4/3/3/5/5/5 = 34	3/5/3/3/2/5/5/5 = 31	rank 3
constant-parameter $K = 4$ hash ensemble	4/5/4/3/3/4/5/5 = 33	3/4/4/2/2/4/5/5 = 29	conditional child
attention post- c_{proj} Peri-RMSNorm	3/5/5/4/4/5/5/5 = 36	2/5/3/2/2/4/5/4 = 27	conditional final round
full Engram key-query contextual gate	5/5/4/4/3/3/5/5 = 34	4/5/3/2/2/2/5/4 = 27	defer to next paper
sidecar retry	—	1/5/1/2/1/1/5/1 = 17	hard veto
canonical-key retry	—	1/5/4/1/2/5/5/2 = 25	hard veto

IDs/targets/bytes, packer boundaries, RNG identity, and compiler identity must match exactly. The new gate gradients must be finite and nonzero. All `PACKER_*` diagnostic flags are zero. A mismatch is `INVALID_DIAGNOSTIC`; it cannot be repaired by relaxing equality after seeing the value.

Applicability mediator. The first execution after replay is one eager, training-only 256-step treatment/control mediator pair; residual probes require compilation to be disabled. Its control rows 1–32 define post-MLP residual variance at layer ℓ as

$$v_{\ell,t} = \text{mean}_{b,s} \text{Var}_d[z_{\ell,t}(b, s, d)],$$

computed in FP64 outside the scored endpoint. In at least 24 of 32 control steps, Spearman correlation between layer index and $\log v_{\ell,t}$ must be at least 0.60 and $v_{7,t}/v_{0,t} \geq 1.25$. Otherwise the local model lacks the source mechanism and the round is killed before endpoint work. Post-attention statistics are recorded but cannot replace the registered post-MLP statistic.

Treatment mediation and safety. Across that eager 256-step pair, the treatment/control ratio of v_7/v_0 must be at most 0.85 and the treatment’s deepest-layer variance must be at most 0.90 of control. Setting the learned gates to zero at the assay endpoint must restore at least 50% of the variance-ratio gap, tying the effect to GPAS rather than incidental drift. Every $1 - \text{SiLU}(\alpha_\ell)$ must stay finite in $[0.5, 1.5]$; all gate and pre-existing gradients must remain finite; and the deepest four layers must contribute at least half of the aggregate variance-ratio reduction.

Only after the eager pair passes, two separate no-probe compiled 256-step pairs run on the same two frozen H200 UUIDs with roles swapped. Every arm in each placement must have exactly one Dynamo graph, zero recompiles, and zero CUDA-graph recordings; timing uses steps 33–256. The counterbalanced

treatment/control token-rate point ratio must be at least 0.99 with a one-sided 95% lower bound of at least 0.985. Each arm’s clean steps 33–256 are partitioned into 28 ordered, nonoverlapping eight-step blocks. With `random.Random(19019063)`, each of 10,000 replicates draws 28 block indices with replacement independently per placement, reusing that placement’s index vector for its control and treatment, and takes the geometric mean of the two aggregate treatment/control token-rate ratios. The lower bound is ascending order statistic 499 (zero based); any missing, nonfinite, or nonpositive block or replicate invalidates the diagnostic. This bootstrap specification was corrected prospectively before any Round-1 GPU execution. Thus one registered seed/pair has exactly three governed pair executions—one eager mediator and two clean timing placements—not three endpoint-funnel pairs. Replay is judged before the eager pair, mediation immediately afterward, and later phases are skipped on any failure. This separation was corrected prospectively before any Round-1 GPU execution. A pass authorizes only drafting and registering a separate ordinary paired endpoint proposal. The predicted raw paired endpoint effect is $\Delta\text{bpb} \leq -0.0030$; ordinary promotion still requires the registered -0.002396 magnitude and confidence gates.

5. Rounds 2–3: Hash Independence

Source and local extrapolation. Engram uses layer/head-specific odd mixers and distinct prime table moduli (Cheng et al., 2026; DeepSeek-AI, 2026). OPHIS already uses disjoint multiplier tuples but applies the same composite modulus $8192 \times 64 = 524288$ to all fourteen active bigram/trigram tables. Round 2 changes only these moduli while preserving raw token IDs; canonicalized IDs are forbidden. Prime bracketing around 524288 is a local constant-sum extrapolation, not an Engram result.

Distinct-prime reserve: definition v1 withdrawn. The proposer score was 35/40 and the earlier critic score was 31/40. After implementation audit, the authoritative critic vector is $4/4/2/2/2/3/2/5 = 24/40$, with verdict **hold**. The first proposed capsule serialized a width of 256 and claimed 1,879,048,192 active memory parameters. Direct inspection instead gives `n_embd`=768, six KV heads, `kv_dim`=768, and a $K = 2$ half-width of 384, so the same 7,340,032 rows contain 2,818,572,288 active parameters. Its stated canonical payload was also not independently reproducible from the initial stated serialization. These are prospective design failures, not experiment results. V1 is **WITHDRAWN_INVALID_SPECIFICATION**; no collision, throughput, BPB, adoption, or mechanism conclusion may be drawn from it.

Round 2 remains a research slot but has no execution authority until a separately versioned, append-only v2 is frozen. V2 must bind exact canonical JSON bytes and an independently reproduced digest; live source and config hashes; the complete fourteen-table inventory; runtime-derived width; parameter membership, shapes, dtypes, counts, and bytes; and the actual optimizer state keys, tensor shapes, dtypes, counts, and bytes under the bound `NGRAM_STATE_ROWWISE` and `NGRAM_SPARSE_GRAD` flags. Arithmetic equality alone is insufficient. V1 remains preserved and cannot be overwritten or used as v2 evidence. If v2 is not frozen when this slot is reached, Round 2 is skipped and Round 3 remains blocked. The replacement may not alter the following metrics after observing corpus values.

Frozen offline census. Aggregate raw n-gram frequency counters from exactly eight pinned training shards; validation data are forbidden. For every order n , layer ℓ , head k , and unordered distinct key pair $g < g'$, let $C_{n\ell k}(g, g')$ indicate a bucket collision and weight it by $f_g f_{g'}$. The double collision is $J_{n\ell} = C_{n\ell 0} C_{n\ell 1}$. The authoritative aggregate collision lift is

$$L_n^w = \frac{Q_n^w \sum_{\ell} D_{n\ell}^w}{\sum_{\ell} B_{n\ell 0}^w B_{n\ell 1}^w}, \quad Q_n^w = \sum_{g < g'} f_g f_{g'}.$$

The control premise requires unweighted expected support at least 100 and a weighted lift point estimate and one-sided 95% bootstrap lower bound at least 1.25. The treatment must have lift at most 1.10 and at most 0.80 of control; weighted double-collision burden must be at most 0.75 of control with one-sided 95% upper bound at most 0.90; unweighted burden must be at most 0.80; and at least six of seven layer

sites may not regress.

Marginal weighted collision burden must be at most 1.005 of control overall and 1.01 in every layer. Repeated full aliases across layer pairs must be at most 0.85 of control. Occurrence-weighted Rényi-2 joint entropy deficit must be at most 0.80 of control, while no individual marginal entropy may fall by more than 5×10^{-4} . The paired bootstrap uses whole-shard resampling, seed 19019063, 10,000 replicates, and the same eight-index vector for both arms. Zero denominators or any non-finite replicate fail closed. This offline pass is necessary but never sufficient for adoption.

The runtime treatment must preserve exact total embedding parameters, optimizer-state elements, data, targets, byte counts, ordering, BOS positions, boundaries, non-memory initialization, and global RNG. It must compile to one graph with zero recompiles and slow charged token rate by at most 0.5%. Its predicted endpoint effect is $\Delta\text{bpb} \leq -0.0030$.

Conditional $K = 4$ child. Round 3 is eligible only after full Round-2 acceptance. It uses four distinct prime tables per injection, each with one quarter of the accepted $K = 2$ concatenated width. Total embedding parameters and every RMSProp state element must match exactly; initial concatenated memory-output variance must be within 2%; non-memory parameters, data, and RNG must be exact. Its joint collision burden must be at most 0.75 of the accepted $K = 2$ comparator on the same census, graph count must be unchanged, and slowdown must be at most 1.0%. The predicted paired effect is $\Delta\text{bpb} \leq -0.0026$. Engram uses eight hash heads; $K = 4$ is a local intermediate, not source-faithful replication.

6. Rounds 4–5: Attention Transport

Conditioned Q/K/V initialization. Round 4 uses the ICLR 2026 paper’s joint construction: independent semi-orthogonal per-head Q/K maps and rectangular-identity V maps (Saratchandran & Lucey, 2026). It does not revive Paper 019’s disjoint-V extrapolation. Before training, each Q/K row-orthogonality residual must be at most 10^{-5} , every V map must satisfy $\sigma_{\min}/\sigma_{\max} \geq 0.99$, and all non-Q/K/V tensors plus global RNG must match control exactly. The registered attention-Jacobian surrogate must fall at least 20% in every layer, while attention entropy changes by at most 0.05.

Because OPHIS initializes $c_{\text{proj}} = 0$, the first backward must produce a finite nonzero c_{proj} gradient, and

Table 2. Frozen order. A failed conditional round is skipped, never replaced post hoc. A substitute requires a new paper.

round	mechanism	dependency
1	GPAS shared gate	local applicability
2	balanced prime moduli	corrected authority
3	constant-parameter $K = 4$	Round 2 accepted
4	conditioned Q/K/V init	none; cumulative R
5	attention Peri-RMSNorm	local imbalance

the backward after the first update must produce finite nonzero Q/K/V gradients. One compiled graph and timing within 0.1% are required. The prediction is $\Delta\text{bpb} \leq -0.0030$. The source reports TinyStories 2.39→2.20 and WikiText-103 44.2→42.7 at 472M parameters, but no local seed variance; the surrogate is a bound, not the true condition number.

Conditional attention Peri-RMSNorm. Round 5 first asks whether the current accepted comparator has the proposed problem. Across 32 fixed diagnostic steps, the post- c_{proj} attention branch must have max/min layer RMS at least 1.25 or layerwise coefficient of variation at least 0.20. Otherwise the round is killed offline. If eligible, apply a fixed scale-free RMS normalization only to the attention output after c_{proj} ; the MLP already has an analogous normalization. This is an OPHIS-specific partial placement, not the full learned-scale Peri-LN architecture (Kim et al., 2025).

Step-0 attention output must remain exactly zero. At step 1, each c_{proj} gradient ratio must lie in $[0.5, 2.0]$ and no layer may exceed 2.5 times control. Across the 32-step diagnostic, residual-norm coefficient of variation and max/min layer-gradient ratio must each fall at least 20%. No non-finite value is allowed; slowdown is at most 1.0%; the predicted paired effect is $\Delta\text{bpb} \leq -0.0030$.

7. Execution, Statistics, and Resource Use

Each endpoint follows a one-pair screen, then cumulative $n = 3$, $n = 6$, and $n = 10$ paired same-seed evidence. A stage-1 value is never a belief, adoption, SOTA, or cross-candidate ranking statistic. Full acceptance requires the current registered mean-effect threshold, sign consistency, confidence gate, and the candidate-specific mediator. Role order is counterbalanced, and a fresh control runs concurrently at every stage. Round 1’s three diagnostic executions precede and do not consume this funnel.

Causal isolation overrides occupancy. The coordinator may run independent CPU literature, paper, census, or verification work while a registered GPU as-

say runs, and may occupy multiple tenant-free H200s with independent same-stage pairs. It may not fill free GPUs with unregistered experiments, run a child before its parent, or create competing unsynchronized schedulers. PID/UUID/device watchdogs, source-hash bindings, and the one-to-two-GPU attribution reserve remain mandatory. Analysis performed outside the charged clock must be symmetric or explicitly diagnostic-only.

Before any launch: preserve the primary paper and exact code commit; register paper, atomic claims, evidence, mechanism, hypothesis, intervention, and experiment; freeze the source diff and resolved arm configuration; require all `PACKER_*` flags to be zero; render generated state; run validate, strict audit, setup, and gate checks; inventory tenant-free GPUs; and commit the authority. Any source, setup, or artifact drift returns the proposal to **hold**.

8. Falsification and Non-Claims

The block is successful as research even if no candidate is adopted, provided each failure is attributed to a pre-registered mediator. GPAS can fail because depthwise variance growth is absent, the gate does not mediate it, or the cost exceeds its limit. Prime moduli can fail because the control has no cross-head dependence, independence does not improve, or marginal entropy regresses. Conditioned initialization can fail its spectral surrogate, gradient-onset, or endpoint prediction. Peri-RMSNorm can fail because no branch imbalance exists or normalization amplifies the zero-output path.

No offline metric, mechanism assay, one-pair value, historical chart point, or paper-reported improvement is a local SOTA. A new SOTA requires a terminal governed result whose preregistered comparator, full funnel, evidence update, decision, snapshot, chart point, and reproducibility code all pass. Until then, the best displayed value remains historical experiment 501 at 0.927183, with its original timestamp and its restricted provenance role.

9. Reproducibility Binding

The GPAS source capsule binds official repository commit 31980688f4cbb1b0cff59bca9077e6fc52dab3f0, tree 076a1d81a48f4d03dbd455d4197345480bd818bb, deterministic source archive SHA-256 1177e6766f53cf0bc59b880a96ca09d7732984db62f8d54b9bf5083b1b45acf0, Git bundle SHA-256 8e2abb5ba02b16639d92cb532b3134c5fa4bcce0f43156585fa2e591294edaf1, and proceedings PDF SHA-256 d644ed31e31a693fa889e01655e94cfea1cdb0c312901ee11b001af68b14a35e. The

pre-intervention hashes preserved in Git commit 16d8fae959b67f45ed1f6338aaa554a6fbb91f07 are:

train.py
e2d4ac9e91c6ef716a8fdab35ab72da3e0e4bb23cbbecee04cc528b66d172b8b

lib.py
13de28bcb398866be65cfca489413818d75dc2fcd2461e367be06198f879123d

research/setup/reconciliation.json
526dcdf38f90c83ca097776c69ec2d9610c6f39091c307f1d60f8403ce698acd. The invalid sidecar pair artifact is preserved with SHA-256 4c26aa74b020b093740e099abc50510be23cd9d353789aee049f42bcebaa4dca; its preservation prevents it from being rediscovered or re-labeled.

10. Conclusion

Paper 020 replaces plateau knob-search with five falsifiable mechanism questions. Its first question is deliberately local: does OPHIS even exhibit the depthwise residual-variance mechanism that GPAS is designed to correct? Its second asks whether existing n-gram heads provide independent evidence rather than correlated aliases, but remains blocked after the implementation auditor caught a stale width and incomplete equality contract. Later rounds condition and scale the attention path only through measured applicability. The accepted-only chain turns separate architectural ideas into one connected body while keeping every causal increment identifiable.

References

- GPT Research Coordinator et al. Paper 019a: Observe Before Optimizing—An Attribution Addendum for Packer-Sourced FA3 Boundaries. Internal OPHIS preregistration, 2026.
- Tianhao Chen, Xin Xu, Zijing Liu, Pengxiang Li, Xinyuan Song, Ajay Kumar Jaiswal, Fan Zhang, Jishan Hu, Yang Wang, Hao Chen, Shizhe Diao, Shiwei Liu, Yu Li, Lu Yin, and Can Yang. GPAS: Accelerating Convergence of LLM Pretraining via Gradient-Preserving Activation Scaling. NeurIPS, 2025.
- Chen et al. GPAS official implementation, commit 31980688f4cbb1b0cff59bca9077e6fc52dab3f0. <https://github.com/dandingsky/GPAS>, captured 2026.
- Xin Cheng, Wangding Zeng, Damai Dai, Qinyu Chen, Bingxuan Wang, Zhenda Xie, Kezhao Huang, Xingkai Yu, Zhewen Hao, Han Zhang, Yu-Kun Li, Huishuai Zhang, Dongyan Zhao, and Wenfeng Liang. Conditional Memory via Scalable Lookup: A New Axis of Sparsity for Large Language Models. ACL, 2026.
- DeepSeek-AI. Engram official demonstration, commit fb7f84a21f91223715394a33a1dc24bbfb7f788e. <https://github.com/deepseek-ai/Engram>, captured 2026.
- Hemanth Saratchandran and Simon Lucey. Conditioned Initialization for Attention. ICLR, 2026.
- Jeonghoon Kim, Byeongchan Lee, Cheonbok Park, Yeontaek Oh, Beomjun Kim, Taehwan Yoo, Seongjin Shin, Dongyoon Han, Jinwoo Shin, and Kang Min Yoo. Peri-LN: Revisiting Normalization Layer in the Transformer Architecture. ICML, 2025.